

Linux Task 03_Nimitha: Process Management, System Monitoring & Basic Shell Scripting

Objective The purpose of this task is to understand how Linux manages processes, monitors system resources, and automates tasks using shell scripts. These are essential skills for Linux Administrators, SOC Analysts, and Cyber Security Professionals. Part A: Process Monitoring

Part F: Security Monitoring Challenge

netstat

ss

who

w
l

ast

Command	Purpose	Example Output	Security Use Case
netstat	Shows network connections	TCP connections list	Detect suspicious network activity
ss	Displays socket statistics	Active TCP/UDP sessions	Monitor network communication
who	Shows logged-in users	username pts/0	Detect unauthorized logins
w	Shows logged-in users and activity	User, login time, command	Monitor user actions
last	Displays login history	Login records	Investigate security incidents

Part G: Mini SOC Activity

1. How would you identify resource-heavy processes?

I would use commands such as `top`, `htop`, and `ps aux` to view CPU and memory usage. Processes using unusually high resources would be investigated further.

2. How would you determine whether a process is suspicious?

I would check the process name, PID, resource usage, owner, and command path. Unknown processes or processes using excessive resources may require further analysis.

3. What information would you collect before terminating a process?

I would collect the PID, process name, CPU usage, memory usage, user account, start time, and purpose of the process. This helps avoid accidentally stopping an important system service.